

Functions in R

Dr. Mutua Kilai, PhD

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Department of Pure and Applied Sciences

Introduction

- Functions allow you to automate common tasks in a more powerful and general way than copy-and-pasting.

Advantages of Functions

- Writing a function has four big advantages over using copy-and-paste:
 - i. You can give a function an evocative name that makes your code easier to understand.
 - ii. As requirements change, you only need to update code in one place, instead of many.
 - iii. You eliminate the chance of making incidental mistakes when you copy and paste
 - iv. It makes it easier to reuse work from project-to-project, increasing your productivity over time.

Contents of a Function

- **A name:** you specify the function name
- **Arguments:** The arguments are things that vary across calls
- **Body** The body is the code that's repeated across all the calls.
- **Return Value** The value returned by the function.
- You create a function using the following template

```
1 name <- function(arguments) {  
2   body  
3 }
```

Vector Functions

- Functions that take one or more vectors and return a vector result.

```
1 # create a vector of age
2
3 age <- c(23,34,23,12,19)
4
5 # create a function that rescales the vector as
6 # age-min(age)/max(age)-min(age)
7
8 age.function <- function(x)
9 {
10   value <- (x-min(x))/(max(x)-min(x))
11   return(value)
12 }
13 # call the function
14
15 age.function(age)
```

Data Frame Functions

- Data frame functions work like dplyr verbs: they take a data frame as the first argument, some extra arguments that say what to do with it, and return a data frame or a vector.

```
1 library(dplyr)
2 library(ggplot2)
3 count_prop <- function(df, var, sort = FALSE) {
4   df |>
5     count({{ var }}, sort = sort) |>
6     mutate(prop = n / sum(n))
7 }
8 diamonds |> count_prop(clarity)
```

```
# A tibble: 8 × 3
  clarity      n  prop
  <ord>    <int> <dbl>
1 I1         741 0.0137
2 SI2        9194 0.170
3 SI1       13065 0.242
4 VS2       12258 0.227
5 VS1        8171 0.151
6 VVS2        5066 0.0939
7 VVS1        3655 0.0678
8 IF         1790 0.0332
```


Plot Functions

Instead of returning a data frame, you might want to return a plot. Fortunately, you can use the same techniques with `ggplot2`, because `aes()` is a data-masking function.

```
1 library(dplyr)
2 library(ggplot2)
3 conditionalBars <- function(df, condition, var) {
4   df |>
5     filter({{ condition }}) |>
6     ggplot(aes(x = {{ var }})) +
7     geom_bar()
8 }
9 diamonds |> conditionalBars(cut == "Good", clarity)
```